

Prognostic Factors for Metastasis Following Proton Beam Irradiation of Uveal Melanomas

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Abstract: Prognostic indicators for the development of metastasis following proton beam irradiation of uveal melanomas were evaluated for 510 patients treated from 1975 to 1984. Thirty-three patients developed metastasis (6.5%) from 3 to 51 months following treatment. The primary site of metastasis was the liver in 28 cases (85%). Both demographic and clinical factors were considered. The three leading predictors of survival without metastasis after proton beam irradiation in order of importance were: (1) largest diameter of the tumor; (2) location of the anterior margin of the tumor; and (3) age at treatment. Worse prognosis was associated with largest tumor diameter greater than 15.0 mm, tumor involvement of the ciliary body and age at treatment older than 59 years. [Key words: metastases, prognostic factors, proton beam irradiation, uveal melanoma.] *Ophthalmology* 93:675-680, 1986

Enucleation was the standard method for treating intraocular melanomas for many years but alternative forms of treatment have been used with increased frequency in recent years.¹⁻⁷ Radiation therapy is the most widely used conservative method and, in spite of the relative radioresistance of these tumors, its effectiveness in destroying uveal melanomas has been demonstrated.³⁻⁷ We have used proton beam irradiation for the treatment of uveal melanomas at the Harvard Cyclotron since 1975, and our results were reported recently.⁸ Although proton beam irradiation is effective in maintaining vision and saving

eyes in many cases metastases do still occur among treated patients. The factors that predict which tumors will metastasize after proton beam irradiation are unknown. We report herein the prognostic indicators for metastasis following proton beam irradiation of uveal melanomas among patients treated from 1975 to 1984. We used multivariate techniques to identify which clinical variables are the leading independent predictors of the outcome of uveal melanoma patients treated by proton beam irradiation.

PATIENTS AND METHODS

PATIENTS

All patients who underwent proton beam irradiation between July 1975 and June 1984 were identified and follow up-data were analyzed up to December 31, 1984. Of 521 patients (522 eyes) treated, 510 are the subject of this study. Of the remaining 11, 7 patients had no follow-

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